How many samples does each question need?

HotpotQA • 500 matched training questions • Four weak/strong combinations • Lenient EM

01 RETRIEVE

Once with W and once with S
W = Qwen2.5-1.5B Instruct; S = 7B Instruct

02 SAMPLE + FUSE

40 answer samples for each model pair
Fuse the first k samples for every k = 1…40

03 SCORE

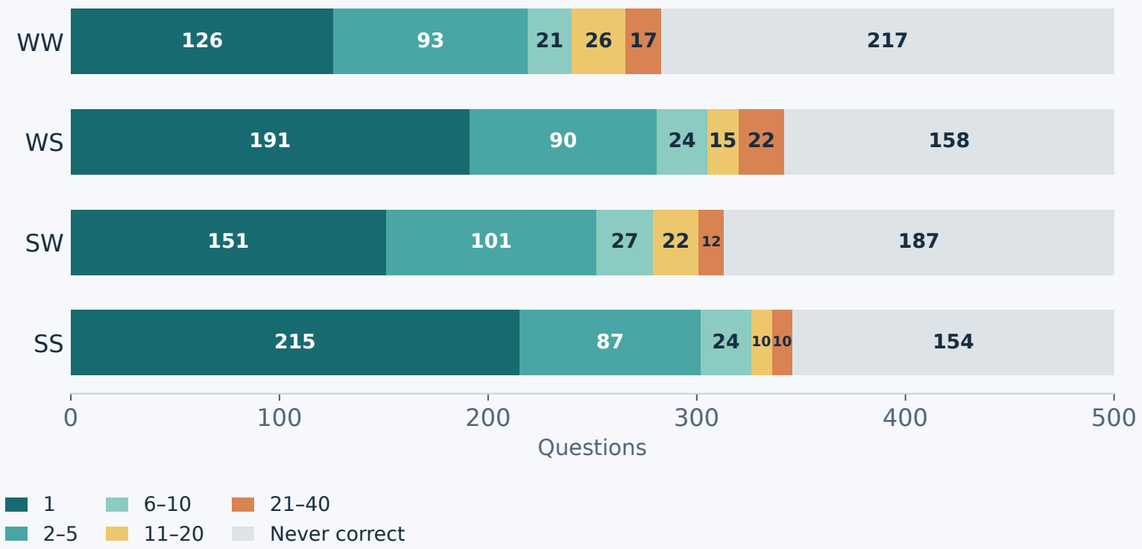
Fixed, gold-blind answer extraction
Compare each result with gold answers

GGUF Q4_K_M on serverA • AgentTTS public HotpotQA prompts • Temperature: retrieval 0.9 / answer 0.7 / fusion 0.3 • Max output 512 tokens

Pair code: retrieval model → answer/fusion model. For example, WS = weak retrieval + strong answer/fusion.

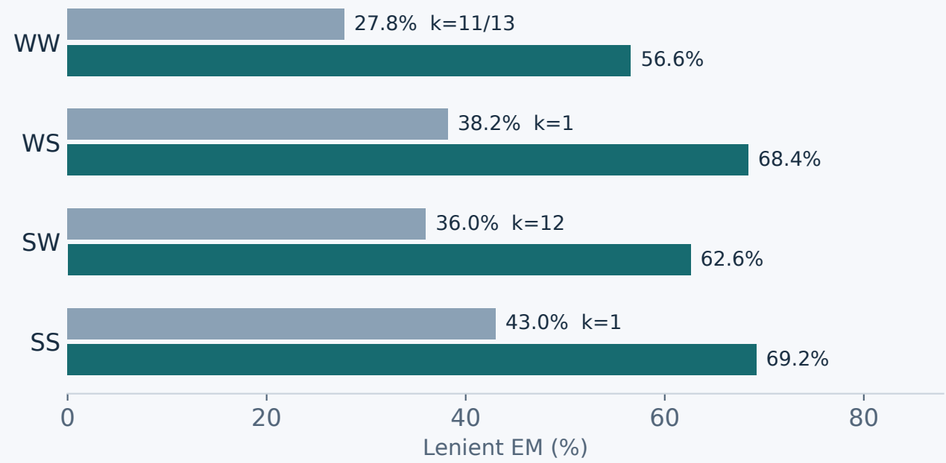
A Earliest successful sampling budget

Question counts; each row totals 500.



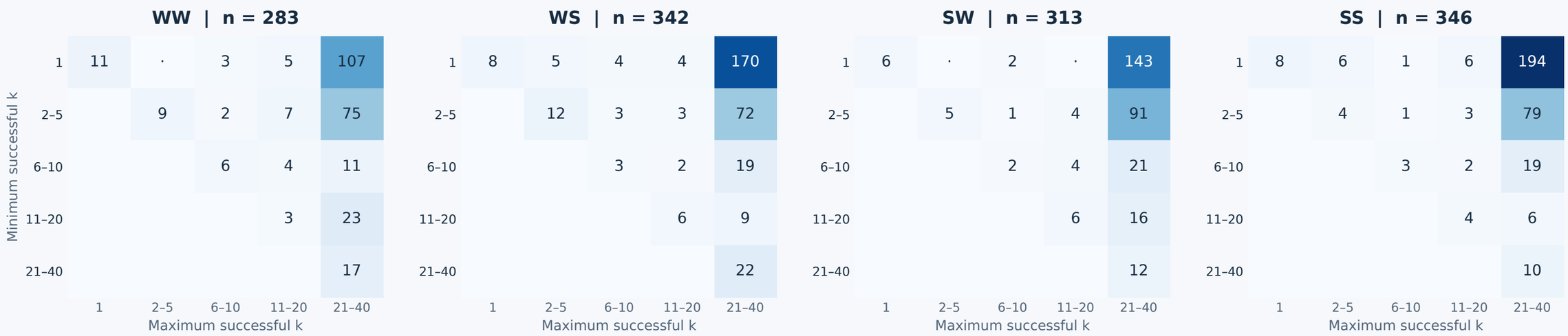
B Fixed versus per-question budgets

Both selected retrospectively on the same 500 questions.



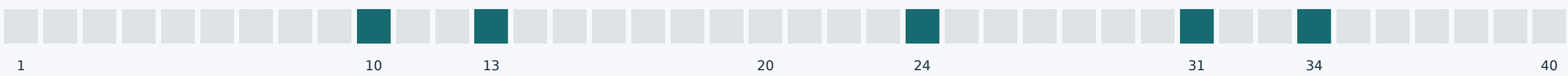
C Where do correct sampling budgets begin and end?

Each heatmap counts questions by their minimum and maximum successful k. Only questions correct at least once are included.



Correctness can have gaps within these bounds: WW 228/283 • WS 264/342 • SW 263/313 • SS 294/346 questions have non-contiguous success sets.

Example One SS question succeeds only at k = 10, 13, 24, 31, 34



Reading the result: extra samples sometimes rescue an answer, but can also change a correct fusion into an incorrect one.

Scope: availability-based training cohort, not a held-out benchmark or an AgentTTS reproduction score. Oracle uses gold answers and is not a deployable or cost-matched policy.

Lenient EM: normalized bidirectional whole-word containment with inequality guards, applied to extracted answers. Local scoring rule; may accept partial matches and differ from the original scorer.

Retrieval is sampled once per model here. Sampling budgets k vary only for answer generation and fusion. Frozen cohort results completed 17 Oct 2026.